Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester Theory B. Tech (All branches) Examination

April-May 2018

ME 104 Basics of Mechanical Engineering

Date: 07.05.2018, Monday **Time: 01:30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1.a Differentiate work and heat. **[02]**

Q – 1.b Explain zeroth law of thermodynamics. **[02]**

Q – 1.c Discuss about various sources of energy. **[02]**

Q – 1.d Represent the constant volume compression and constant pressure compression process on p-V diagram with the same initial conditions. **[03]**

Q – 2 **Answer any three.** **[12]**

1. Explain dry steam and superheated steam with its enthalpy equations.

2. Enlist various refrigerants and give applications of refrigeration.

3. Determine enthalpy of 1 kg of steam at a pressure 12 bar when
(i) steam is dry and saturated
(ii) steam is 22% wet and
Take C_{ps} = 2.1 kJ/kg K.

Pressure (Bar)	**Saturation temperature (°C)**	**Enthalpy of Liquid (kJ/kg)**	**Enthalpy of Evaporation (kJ/kg)**
12	188.0	798.4	1984.3

4. Explain window air conditioning system with neat sketch.

Q – 3 **Answer any two.** **[14]**

1. Explain vapour absorption refrigeration system with neat sketch.

2. Differentiate fire tube boiler and water tube boiler.

3. Derive combined gas law.

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SECTION – II

Q – 4.a Explain spur gear. **[02]**

Q – 4.b What do you mean by Otto cycle engine and Diesel cycle engine? **[02]**

Q – 4.c Enlist applications of compressor. **[02]**

Q – 4.d Differentiate open belt drive and cross belt drive. **[03]**

Q – 5 **Answer any three.** **[12]**

1. Classify air compressor.
2. Explain rigid couplings and flexible couplings with their examples.
3. Draw line diagram of an I.C. engine. Show different parts of it.
4. How external combustion engine differs from internal combustion engine?

Q – 6 **Answer any two.** **[14]**

1. A 6 cylinder 4 stroke I.C. engine is to produce 95kW brake power at 800 rpm. The stroke to bore ratio is 1.25. Mean effective pressure is 7 bar. Determine the bore and stroke of the engine. Assume mechanical efficiency as 80%.
2. Explain double block brake with neat sketch.
3. Explain double acting reciprocating pump with neat sketch.

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